Zhichao "Carton" ZENG

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UAT keywords: Dark matter | Galaxy dark matter halos | N-body simulations

Professional Experience

Texas A&M University — College Station, Texas, USA
Postdoctoral Associate — *9/2024-present*
- PI: Louis Strigari

Education

The Ohio State University — Columbus, Ohio, U.S.
Ph.D. in Physics — *8/2018 – 8/2024*
- Thesis: Self-interacting dark matter subhalos and dwarf satellite galaxies: evolution and diversity
- Advisor: Annika H. G. Peter

The Chinese University of Hong Kong — Hong Kong
M.Phil. in Physics — *8/2016 – 6/2018*
- Thesis: Effects of cosmological neutrinos on the large-scale structure formation
- Advisor: Ming-Chung Chu

The Chinese University of Hong Kong — Hong Kong
B.Sc. in Physics — *9/2012 – 6/2016*
- Thesis: Influence of massive and degenerate neutrinos on the evolution of the Universe
- Advisor: Ming-Chung Chu

Research Interests

Theoretical/computational modelling: dark matter, galactic dynamics, halo–subhalo dynamics, structure formation
Observational probes: dwarf (satellite) galaxies, strong lensing, stellar streams, massive black holes
Methods and tools: N-body simulation (Arepo, Gadget2, Amiga halo finder), semi-analytic/hybrid models

Publications & Pre-prints

Total Publications (15) — *Total Citations: 479*

Leading Author Publications (6) — **Total Citations: 176**

[1] **Z. C. Zeng**, A. Peter, L. Strigari, A. Benson, X. Du, C. Mace, *Feeding the Void: Co-evolution of the SIDM-seeded Black Hole and the dark halo after core-collapse*, 2026, arXiv:2609.22758. *cited by: 0*

[2] **Z. C. Zeng**, O. Hartl, L. Strigari, A. Peter, X. Du, C. Mace, A. Benson, *Bypassed Core Formation in Milky Way-Mass SIDM Halos: Implications for the Local Group Past-Pericenter Scenario*, 2026, Phys.Rev.D, arXiv:2604.08647. *cited by: 2*

[3] **Z. C. Zeng**, A. Peter, X. Du, A. Benson, J. Li, et al., *Diversity and universality: evolution of dwarf galaxies with self-interacting dark matter*, 2025, Phys.Rev.D, arXiv:2412.14621. *cited by: 14*

[4] **Z. C. Zeng**, A. Peter, X. Du, S. Yang, A. Benson, et al., *Till the core collapses: the evolution and properties of self-interacting dark matter subhalos*, 2025, Phys.Rev.D, arXiv:2310.09910. *cited by: 32*

[5] **Z. C. Zeng**, A. Peter, X. Du, A. Benson, S. Kim, et al., *Core-collapse, evaporation and tidal effects: the life story of a self-interacting dark matter subhalo*, 2022, MNRAS, arXiv:2110.00259. *cited by: 108*

[6] **Z. C. Zeng**, S. Yeung and M.-C. Chu, *Effects of neutrino mass and asymmetry on cosmological structure formation*, 2019, JCAP, arXiv:1808.00357. *cited by: 20*

Mentored Student Publications (3) **Total Citations: 63**

[1] C. Mace, B. Dhanasingham, **Z. C. Zeng**, et al., *The Sensitivity of Substructure Lensing to SIDM Core-collapse Model Variation*, 2026, submitted to Phys.Rev.D, arXiv:2605.24174. *cited by: 3*

[2] C. Mace, S. Yang, **Z. C. Zeng**, et al., *Calibrating the SIDM Gravothermal Catastrophe with N-body Simulations*, 2026, Phys.Rev.D, arXiv:2504.13004. *cited by: 15*

[3] C. Mace, **Z. C. Zeng**, A. Peter, et al., *Convergence Tests of Self-Interacting Dark Matter Simulations*, 2024, Phys.Rev.D, arXiv:2402.01604. *cited by: 45*

Collaborative Publications (6) **Total Citations: 240**

[1] A. Engelhardt, et al. (inc. **Z. C. Zeng**), *MARVELously Dark: the density profile evolution of dwarf halos in velocity-dependent SIDM*, 2026, JCAP, arXiv:2601.23264. *cited by: 8*

[2] X. Du, A. Benson, **Z. C. Zeng**, et al., *Tidal evolution of cored and cuspy dark matter halos*, 2024, Phys.Rev.D, arXiv:2403.09597. *cited by: 36*

[3] S. Yang, et al. (inc. **Z. C. Zeng**), *A quantitative comparison between velocity dependent SIDM cross-sections constrained by the gravothermal and isothermal models*, 2024, MNRAS, arXiv:2305.05067. *cited by: 28*

[4] S. Yang, X. Du, **Z. C. Zeng**, A. Benson, F. Jiang, E. O. Nadler, A. Peter, *Gravothermal Solutions of SIDM Halos: Mapping from Constant to Velocity-dependent Cross Section*, 2023, ApJ, arXiv:2205.02957. *cited by: 86*

[5] F. Jiang, et al. (inc. **Z. C. Zeng**), *A semi-analytic study of self-interacting dark-matter haloes with baryons*, 2023, MNRAS, arXiv:2206.12425. *cited by: 64*

[6] R. Emami, et al. (inc. **Z. C. Zeng**), *Evidence of Neutrino Enhanced Clustering in a Complete Sample of Sloan Survey Clusters, Implying* $\sum m_\nu = 0.119 \pm 0.034$ eV, 2017, arXiv:1711.05210. *cited by: 18*

PRESENTATIONS

Self-interacting dark matter subhalos and dwarf satellite galaxies

- Poster, Galactic Frontiers II, Dartmouth College — June 2025
- Poster, Gravity in the Local Group, CMU — June 2025
- Plenary talk, Theoretical Astroparticle and Cosmology Symposium in Texas, Texas A&M — Oct 2024
- Texas A&M Astro Seminar — Sept 2024
- Plenary talk, Dark Matter 2024, KITP, UCSB — Jun 2024
- Ohio State HEP Seminar — Mar 2024

Till the core collapses: evolution of SIDM subhalos

- Plenary talk, Galactic Frontiers, CCA Flatiron Institute — July 2023
- Seminar (virtual), Nature of Dark Matter, Yale — June 2023

Core-collapse, evaporation and tidal effects: the life story of a self-interacting dark matter

- Plenary talk (virtual), N-body-shop workshop, CCA Flatiron Institute — June 2022
- Seminar (virtual), The McWilliams Center for Cosmology, CMU — Apr 2022
- Poster (virtual), COSMO21 — July 2021

MENTORING

Steven Zhu (Undergraduate Student at OSU & UMich)	2025 summer
Odelia Hartl (Graduate Student at TAMU)	2024 – 2026
Chloe Zheng (Graduate Student at OSU)	2024 – Present
Angelica Whisnant (Graduate Student at OSU)	2024 – Present
Charles Mace (Graduate Student at OSU)	2021 – 2026
Jiashen He (Undergraduate Student at OSU)	2019 summer

Teaching Experience

Graduate Teaching Assistant 2018, 2019, 2022
The Ohio State University *Columbus, OH*

- Physics courses: Mechanics, Fluids, Waves (Introductory Physics for Engineers and Physicists).
- Astronomy courses: Introduction to Astronomy (General Education) — grading and exam writing; Stellar Evolution — grading.
- Led recitation and laboratory sections, and graded coursework, for undergraduate students.

Graduate Teaching Assistant 2016 – 2018
The Chinese University of Hong Kong *Hong Kong*

- Physics courses: Thermodynamics, Computational Physics, General Physics.
- Led recitation and laboratory sections, and graded coursework, for undergraduate students.

Awards & Honors

OSU Presidential Fellowship (Graduate School Dissertation Year Fellowship, $36k+tuition) 2023 – 2024
Professor Charles K. Kao Research Exchange Scholarship ($5k) 2013 – 2014
University Full Admission Scholarship (CUHK, $25k+tuition) 2012 – 2016

Professional Service

Referee: The Astrophysical Journal, Physical Review Letters, European Physical Journal C